

Homework Problems. Solutions should be written/typed in numerical order and clearly labeled. Unsolved problems should be included but left blank.

Use the following to answer the next two problems. Let $E = \{1, 2, 3\}$. Consider the following relations in E .

$$\mathcal{R}_1 = \{(1, 1), (2, 1), (2, 2), (3, 2), (2, 3)\} \quad \mathcal{R}_4 = \{(1, 1), (3, 2), (2, 3)\}$$

$$\mathcal{R}_2 = \{(1, 1)\} \quad \mathcal{R}_5 = E \times E$$

$$\mathcal{R}_3 = \{(1, 2)\}$$

1. State whether each relation above, $\mathcal{R}_1, \mathcal{R}_2, \mathcal{R}_3, \mathcal{R}_4, \mathcal{R}_5$, is symmetric. Explain your answers.

$\mathcal{R}_1$: No. $(2, 1) \in \mathcal{R}_1$, but $(1, 2) \notin \mathcal{R}_1$ so $\mathcal{R}_1 \not\subseteq \mathcal{R}_1^{-1}$.

$\mathcal{R}_2$: Yes. $(1, 1) \in \mathcal{R}_2$ so $1\mathcal{R}_2 1$.

$\mathcal{R}_3$: No. $(1, 2) \in \mathcal{R}_3$, but $(2, 1) \notin \mathcal{R}_3$ so $1\mathcal{R}_3 \not\subseteq 1$.

$\mathcal{R}_4$: Yes. $(3, 2) \in \mathcal{R}_4$ and $(2, 3) \in \mathcal{R}_4$.

$\mathcal{R}_5$: Yes. $E \times E = \{(x, y) \mid x \in E \text{ and } y \in E\}$, i.e., $E \times E$ is the set of all pairs (x, y) such that $x \in E$ and $y \in E$. So for any $x, y \in E$, if $(x, y) \in E \times E = \mathcal{R}_5$ then $(y, x) \in E \times E = \mathcal{R}_5$.

$$\mathcal{R}_6 = \{(1, 2), (2, 2)\} \quad \mathcal{R}_9 = \{(1, 1)\}$$

$$\mathcal{R}_7 = \{(1, 2), (2, 3), (1, 3), (2, 1), (1, 1)\} \quad \mathcal{R}_{10} = E \times E$$

$$\mathcal{R}_8 = \{(1, 2)\}$$

2. State whether each relation above, $\mathcal{R}_6, \mathcal{R}_7, \mathcal{R}_8, \mathcal{R}_9, \mathcal{R}_{10}$, is transitive. Explain your answers.

$\mathcal{R}_6$: Yes. The transitive condition does not specify that $a \neq b \neq c$.

For $(1, 2)$, let $a = 1$, $b = 2$, and $c = 2$. So for $(1, 2)$ we have $1\mathcal{R}_6 2$ and $2\mathcal{R}_6 2$ and also have $2\mathcal{R}_6 2$ which satisfies the transitive condition. Similarly for $(2, 2)$, let $a = 2$, $b = 2$, and $c = 2$ which gives $2\mathcal{R}_6 2$, $2\mathcal{R}_6 2$, and $2\mathcal{R}_6 2$ which satisfies the transitive condition.

$\mathcal{R}_7$: No. $(2, 1) \in \mathcal{R}_7$ and $(1, 2) \in \mathcal{R}_7$, but $(2, 2) \notin \mathcal{R}_7$ so $2\mathcal{R}_7 \not\subseteq 2$.

$\mathcal{R}_8$: Yes. A relation $\mathcal{R}$ is *transitive* if whenever $a\mathcal{R}b$ and $b\mathcal{R}c$ then $a\mathcal{R}c$. There seems to be confusion when there “is no $b\mathcal{R}c$ relation”. The absence of this case means that the transitive condition is satisfied.

$\mathcal{R}_9$: Yes.

$\mathcal{R}_{10}$: Yes. $E \times E = \{(x, y) \mid x \in E \text{ and } y \in E\}$. So if $(x, y), (y, z) \in E \times E$ for any $x, y, z \in E$ then $(x, z) \in E \times E = \mathcal{R}_{10}$.

3. Define the following relation on the real numbers, $\mathcal{R} : x\mathcal{R}y \iff xy \geq 0$. Prove that $\mathcal{R}$ is reflexive and symmetric, but that $\mathcal{R}$ is not an equivalence relation.

Proof. ($\mathcal{R}$ is reflexive). $x \cdot x = x^2 \geq 0$ for all $x \in \mathbb{R}$. So $\mathcal{R}$ is reflexive.

($\mathcal{R}$ is symmetric). Let $x, y \in \mathbb{R}$. If $xy \geq 0$ then because multiplication is commutative $xy = yx \geq 0$. So $\mathcal{R}$ is symmetric.

($\mathcal{R}$ is not an equivalence relation). To show this, we need to show that $\mathcal{R}$ is not transitive. Consider $1, 0, -1 \in \mathbb{R}$. $1 \cdot 0 \geq 0$ and $0 \cdot (-1) \geq 0$, but $1 \cdot (-1) = -1 \not\geq 0$. So $\mathcal{R}$ is not transitive. $\square$

4. Define the following equivalence relation on the set of subsets of X as follows: $A\mathcal{R}B \iff A \cap B \neq \emptyset$ for $A, B \subseteq X$. Prove or disprove that $\mathcal{R}$ is an equivalence relation.

Proof. $\mathcal{R}$ is not an equivalence relation because it is not reflexive. $\emptyset \subseteq X$ but $\emptyset \cap \emptyset = \emptyset$ so $\emptyset \mathcal{R} \emptyset$.

Note that if the relation was defined on the “set of non-empty subsets of X ”, then it would be reflexive, symmetric, but not transitive. So it still would not be an equivalence relation. $\square$

5. Let $\mathcal{R}$ and $\mathcal{R}'$ be symmetric relations in a set A . Prove that $\mathcal{R} \cap \mathcal{R}'$ is a symmetric relation in A . (note that $\mathcal{R}$ and $\mathcal{R}'$ are subsets of $A \times A$).

Proof. Let $x, y \in A$ such that $(x, y) \in \mathcal{R} \cap \mathcal{R}'$. Since $(x, y) \in \mathcal{R}$, $(x, y) \in \mathcal{R}'$, and both $\mathcal{R}$ and $\mathcal{R}'$ are symmetric it follows that $(y, x) \in \mathcal{R}$ and $(y, x) \in \mathcal{R}'$. Thus $(y, x) \in \mathcal{R} \cap \mathcal{R}'$ so $\mathcal{R} \cap \mathcal{R}'$ is symmetric. $\square$

6. Determine the following:

$$\begin{aligned} \text{a. } \gcd(2^4 \cdot 3^2 \cdot 5 \cdot 7^2, 2 \cdot 3^3 \cdot 7 \cdot 11) \\ = 2 \cdot 3^2 \cdot 7 \end{aligned}$$

$$\begin{aligned} \text{b. The lowest common multiple: } \text{lcm}(2^4 \cdot 3^2 \cdot 5 \cdot 7^2, 2 \cdot 3^3 \cdot 7 \cdot 11). \\ = 2^4 \cdot 3^3 \cdot 5 \cdot 7^2 \cdot 11 \end{aligned}$$

7. Determine $51 \bmod 13$, $342 \bmod 85$, and $(82 \cdot 73) \bmod 7$. (Answers are not unique.)

$$51 = 13 \cdot 3 + 12 \text{ so } 51 \bmod 13 = 12.$$

$$342 = 85 \cdot 4 + 2 \text{ so } 342 \bmod 85 = 2.$$

$$(82 \cdot 73) \bmod 7 = (82 \bmod 7 \cdot 73 \bmod 7) \bmod 7 = (5 \cdot 3) \bmod 7 = 1$$

8. Find integers s and t so that $1 = 7 \cdot s + 11 \cdot t$. Show that s and t are not unique.

Proof. Consider $s = -3$ and $t = 2$. These can be found as follows:

$$11 = 7 \cdot 1 + 4$$

$$7 = 4 \cdot 1 + 3$$

$$4 = 3 \cdot 1 + 1$$

$$3 = 1 \cdot 3 + 0 \text{ so } 1 = \gcd(11, 7)$$

$$1 = 3 \cdot (-1) + 4$$

$$1 = [7 - 4 \cdot 1](-1) + 4 = 7(-1) + 4(2)$$

$$1 = 7(-1) + [11 - 7](2)$$

$$1 = 7(-3) + 11(2)$$

To see that s and t are unique, consider $s = 8$ and $t = -5$. Note that $as + bt = as + [abk - abk] + bt = a(s + bk) + b(t - ak)$ for any $k \in \mathbb{Z}$. Using the answer above, this gives us an infinite class of solutions. For $k = 1$, $7(-3 + 11) + 11(2 - 7) = 7(8) + 11(-5) = 1$. $\square$

Equations of the form $ax + by = d$ are called Diophantine equations (after Diophantus of Alexandria 200-284 AD). Diophantine equations have a solution if and only if the $\gcd(a, b) \mid d$, and if a solution exists then there are infinitely many of them.

9. Let n be a positive integer greater than 1 and let a and b be positive integers. Prove that $a \bmod n = b \bmod n$ if and only if $a = b \bmod n$.

To prove a “**A** if and only if **B**” statement, you must show that **A** implies **B** and **B** implies **A** (“if and only if” is also written “iff.” and $\iff$).

We discussed two equivalent ways to define $a = b \bmod n$:

$$a = b \bmod n \iff n \mid b - a, \text{ and}$$

$$a = b \bmod n \iff a = qn + b \text{ for some } q \in \mathbb{Z}.$$

Proof. ($a \bmod n = b \bmod n \Rightarrow a = b \bmod n$) If $a \bmod n = b \bmod n$ then $n \mid (a - b \bmod n)$ and $n \mid (b - a \bmod n)$. So then $n \mid ([a - b \bmod n] - [b - a \bmod n])$, and since $-b \bmod n + a \bmod n = 0$ the two middle terms cancel out implying that $n \mid (a - b)$. Thus $a = b \bmod n$.

($a = b \bmod n \Rightarrow a \bmod n = b \bmod n$) Suppose that $a = b \bmod n$ so then $n \mid b - a$. Since $n \mid a - a$, $a = a \bmod n$. It follows that $n \mid b - a \Rightarrow n \mid b - a \bmod n \Rightarrow a \bmod n = b \bmod n$. $\square$

(Alternatively)

Proof. ($a \bmod n = b \bmod n \Rightarrow a = b \bmod n$) If $a \bmod n = b \bmod n$ then $a = nq_1 + r$ and $b = nq_2 + r \Rightarrow a - b = n(q_1 - q_2) \Rightarrow n \mid a - b \Rightarrow a = b \bmod n$.

($a = b \bmod n \Rightarrow a \bmod n = b \bmod n$) Suppose that $a = b \bmod n$ then $a = q_1n + r_1$ and $b = q_2n + r_2$. Since $a = b \bmod n \Rightarrow n \mid a - b \Rightarrow n \mid n(q_1 - q_2) + (r_1 + r_2)$. Thus n divides $n(q_1 - q_2)$ and also divides $r_1 + r_2$. Since $0 \leq r_1, r_2 < n$ it follows that $r_1 - r_2 = 0 \Rightarrow r_1 = r_2$. Thus $a \bmod n = b \bmod n$. $\square$

10. Let $d = \gcd(a, b)$. If $a = da'$ and $b = db'$ for some $a', b' \in \mathbb{Z}$, show that $\gcd(a', b') = 1$.

Proof. By the GCD is a linear combination theorem, $d = ax + by$ for some $x, y \in \mathbb{Z}$. This implies that $d = da'x + db'y$, and so dividing by d gives $1 = a'x + b'y$. Thus $1 = \gcd(a', b')$. $\square$